

Topic: Set Operations in SQL

Q. Discuss the SQL Set Operations. Explain UNION, UNION ALL, INTERSECT, and EXCEPT with examples, and list the strict rules required to use them.

Definition to Remember

SQL Set Operations allow you to combine the result sets of two or more independent `SELECT` queries into a single output. They are based on mathematical set theory and include `UNION`, `UNION ALL`, `INTERSECT`, and `EXCEPT/MINUS`. All require **Union Compatibility** between the queries.

1. Rules for Set Operations (Union Compatibility)

Both `SELECT` queries must satisfy:

1. **Same Number of Columns:** Both queries must return the same number of columns.
2. **Compatible Data Types:** Corresponding columns must have compatible data types in the same order.

2. Types of Set Operations

1. UNION

- Combines results of two queries and **automatically removes duplicate rows**.

```
SELECT City FROM Customers
UNION
SELECT City FROM Suppliers;
```

Result: All unique cities from both tables.

2. UNION ALL

- Combines results but **retains all duplicates**. Significantly **faster** than `UNION` (no sorting/deduplication step).

```
SELECT City FROM Customers
UNION ALL
SELECT City FROM Suppliers;
```

Result: All city entries, including repeated cities.

3. INTERSECT

- Returns only rows **common to both** queries.

```
SELECT City FROM Customers
INTERSECT
SELECT City FROM Suppliers;
```

Result: Only cities with both a Customer AND a Supplier.

4. EXCEPT / MINUS

- Returns rows from the **first query that are NOT in the second**.
- Oracle uses `MINUS` ; SQL Server/PostgreSQL use `EXCEPT` .

```
SELECT City FROM Customers
EXCEPT
SELECT City FROM Suppliers;
```

Result: Cities with Customers but NO Suppliers.

3. Quick Comparison

Operation	Duplicates	Returns
UNION	Removed	All from A + All from B
UNION ALL	Kept	All from A + All from B (with duplicates)
INTERSECT	Removed	Common rows only
EXCEPT/MINUS	Removed	Rows in A not in B

★ Must-Write Points (for 10 marks)

1. Set operations combine results of two SELECT queries into a single result set.
2. **Union Compatibility** required: same number of columns AND compatible data types.
3. `UNION` : combines all rows, **removes duplicates**.
4. `UNION ALL` : combines all rows, **keeps duplicates** (faster than UNION).
5. `INTERSECT` : returns only **common rows** present in both queries.
6. `EXCEPT` (MINUS in Oracle): returns rows in first query **not found** in the second.
7. UNION ALL is preferred for performance when duplicates are acceptable.

⚡ Quick Recall

Set Ops → Union Compatibility → UNION (no duplicates) → UNION ALL (keeps duplicates, faster) → INTERSECT (common only) → EXCEPT/MINUS (A not in B)